SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

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Ein Overlay für soziale Netzwerke auf der Grundlage gerichteter azyklischer Graphen auf der Kante

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I confirm that this master's thesis in informatics is my own work and I have documented all sources and material used.

Munich, 15.8.2026 Dan Bachar

Acknowledgments

Abstract

Recent crises, natural disasters, and large-scale protests have highlighted the significant reliance of people on online social networks and instant messaging platforms for communication. During crises, consolidated communication systems can become strained, partially or wholly destroyed, censored, or otherwise monitored. Permissionless distributed overlays are alternatives to consolidated communication realizations. System designers consider these overlays to be more resilient, since they distribute power and content control among many equal peers. These overlays introduce a massive performance overhead, as they rely on gossip and constant chatter between peers to disseminate the correct system state by reaching consensus, through many rounds of communication. This performance overheaf poses a risk to usablity of such systems in disaster zones, as peer devices often have limited connectivity and power supply, which together with the gossip protocol, can severely limit the peers' ability to communicate. In this paper, we propose a system architecture for unstructured peer-to-peer networks that achieves grassroots constitutional consensus based on efficient geometrical overlay design that trade achieve a higher throughput by using more edge computing. We use this system architecture to evaluate a decentralized social network deployed on smartphones using BLE connectivity, where users have total control of their data and privacy.

Kurzfassung

Recent crises, natural disasters, and large-scale protests have highlighted the significant reliance of people on online social networks and instant messaging platforms for communication. During crises, consolidated communication systems can become strained, partially or wholly destroyed, censored, or otherwise monitored. Permissionless distributed overlays are alternatives to consolidated communication realizations. System designers consider these overlays to be more resilient, since they distribute power and content control among many equal peers. These overlays introduce a massive performance overhead, as they rely on gossip and constant chatter between peers to disseminate the correct system state by reaching consensus, through many rounds of communication. This performance overheaf poses a risk to usablity of such systems in disaster zones, as peer devices often have limited connectivity and power supply, which together with the gossip protocol, can severely limit the peers' ability to communicate. In this paper, we propose a system architecture for unstructured peer-to-peer networks that achieves grassroots constitutional consensus based on efficient geometrical overlay design that trade achieve a higher throughput by using more edge computing. We use this system architecture to evaluate a decentralized social network deployed on smartphones using BLE connectivity, where users have total control of their data and privacy.

Contents

1. Introduction

Autocratic regimes have been employing internet shutdowns as an efficient means to suppress democratic protests, leading to fragmentation in the coordination of civil actions and state-sponsored violence [1, 2, 3]. In addition to physical repression, a "networked autocracy" [4], where regimes integrate controls over the media, finance, and information, to oppress legitimate opposition to them without costs to their reputation. Recent crises highlight this trend: during widespread protests in Iran early 2026 in Iran, triggered by the economic collapse of the currency (Rial) leading vendors in the main markets of Tehran to protest, the Islamic Republic has severed international prospects of the ongoing by suppressing access to the Internet, which imposed near-total blackout to create an information vacuum [1]. Similar practices have been utilized to stifle democratic movements globally, including during the 2025 protests in Nepal against a government ban on major social media platforms, the 2026 elections in Uganda, the 2024 student protests in India, and more [4, 5, 6]. In addition to policial repression engaged by the government, physical distruptions can also limit free speech. As evident from communication disruptions due to airstrikes on main fiber optic lines, and the depletion of fuel for generators, information flow was severely limited to the population of Gaza since the start of the Israel-Hamas war in October 2023, halting humanitarian coordination, and leaving the affected population entirely cut off from life-saving information and assistance [7, 8]. Together, the aforementioned practices have led to a growing interest in peer-to-peer (p2p) messaging platforms, decentralized social networks [9, 10], and distributed ledger-based applications [11, 12].

Mobile Ad-Hoc Networks (MANETs) are infrastructure-less networks established using an array of battery-powered mobile devices that use the ad-hoc mode of their wireless network adapter to establish a spontaneous end-to-end connected, packet-switched network [13, 14]. In these networks, nodes act both as routers and hosts for data packets. MANETs, as well as Vehicle Ad-Hoc Networks (VANETs) and Smartphone Ad-Hoc Networks (SPANs), have been extensively studied, with research going back to DARPA's usage of radio to transmit data packets [15]. These elements were bulky, expensive, provided poor data rates, and handled node mobility poorly. Since their onset, MANETs have advanced to VANETs, which deal with rapid topology changes arising due to a network established between fast-moving vehicles such as drone nets [13], and SPANs, which focus on establishing networks around humans utilizing a commonly available resource, the smartphone. SPANs utilize commonly available infrastructure in normal commercial-grade smartphones to connect users without external infrastructure, by leveraging technologies like Wi-Fi Aware, Wi-Fi Direct [16], Bluetooth Classic, and Bluetooth Low Energy [17], to facilitate critical communication during natural disasters, wars, protests, or Internet outages. Because they rely on no external infrastructure, SPANs are oftentimes proposed as the foundation to serverless grassroots social networks such as Bitchat [18], and indeed the idea behind SPANs and is tangential to the core of this research work: MANETs today offer a generally higher data rate than DTNs, but their requirement of a continuously-connected end-to-end path between source and destination make it impossible here, as oftentimes users suffer from offline periods due to emergencies. To route packets, MANETs usually relay on protocols that use topological knowledge of the nodes to efficiently route between them, being reactive to changes or working proactively to evaluate link states, and use out-of-band information like geological location [19]. An alternative to MANETs, DTNs have been proposed in communication-challenged regions where end-to-end path connectivity is lacking: research on DTNs was propelled by NASA for interplanetary travel; the Curiousity rover on Mars, a resource-constrained robot that has been travelling the surface of the red planet since 2012, has been using a DTN to convey pictures to Earth through a satellite orbiting the planet. Instead of wasting precious power on beaming a transmission directly to Earth, the Curiosity rover stores the transmission until the satellite is optimally located orthogonally over it, before sending the satellite its data packets. The satellite then carries the packets until it is at an optimal orbit to Earth. The final transmission between the satellite and Earth completes the communciation cycle, enabling the rover to continue to explore Mars for ten years after the end of its intended two-year mission. The concept that thus lies at the core of DTNs is the Store-Carry-Forward paradigm, which stands in stark contrast to the immediacy requirements of normal packet routing: in standard packet routing, all nodes participating in a transmission between source and destination need to be connected and online concurrently; in DTN routing, the networking graph can be partially offline for extensive periods of time while nodes carry the transmissions. Similarly to MANETs, DTNs also suffer from communication overheads that arise due to control messages, and are especially exacerbated by dense environments with many participants. In addition to space travel, DTNs have also enabled nomadic populace in remote, hard-to-access environments lacking infrastructure the access to communication. The Laponia region near the Arctic Circle, homeland of the Sámi people, is characterized as a sparsely-populated region full of icy fjords, deep valleys, glaciers and mountains. The inhabitants of this region have traditionally been reindeer-herders, and often migrate with the reindeer herds along the yearly cycle. Due to the harsh terrain, sparse level of population, and partial autonomy of the Sámi council, the Sámi people lack reliable cellular and internet infrastructure. To provide the Sámi people with modern communication options, the traditional packet-switching paradigm of communication as conceptualized by the Internet is not optimal. Instead, Lindgren et al. propose the the Store-Carry-Forward paradigm using DTNs [20], where hosts store messages until an opportunity arises to forward them, carry them while moving until they reach a rendezvous with another node or a network attachment point, and then forward them upon contact. This paradigm enables the opportunistic usage of technologies such as Bluetooth to forward messages, which is ubiquitous to virtually all Smartphones and computers today. Insofar as the Internet is not available, DTNs using Store-Carry-Forward networking can be used as a fallback, being supplemented by traditional Internet infrastruture as it becomes available.

To guarantee free speech in disaster zones, during protests, and for citizens living under oppressive regimes, we decide to focus on unstructured p2p networks, as they prove to be more efficient in high-mobility scenarios charcterized by the movement patterns of people, where the network topology is characterized by a high churn rate and a high rate of change, in comparison to structured p2p overlays. Structured p2p networks usually require an expensive randomized approach to establish resilient topologies, as they need to spread information across random peers to prevent dependence on any single peer [21, 22, 23]. Additionally, iterative diffusion of the information causes data plane overhead [22], which increases the message propagation latency [21, 24]. Popular blockchains, which use a structued p2p overlay, employ concentration of services in a few highly available nodes, poses a threat to resilience despite a high reliability of those nodes, as a single network partition can prevent the entire network from communicating [25].

Unstructured p2p networks, on the other hand, are more suitable for highly-dynamic alternative communication methods, often necessary to guarantee free decentralized communication means, as they usually do not distribute central information amongst the participating nodes. Most notably, the rise of unstructured p2p messaging platforms like Firechat [26], Briar [27], bitchat [18], and Amigo [28], have proven that there is a growing shift in user interest patterns away from traditional Centralized Social Networks towards diverse p2p social networks.

After exploring the different types of p2p overlay networks, we discuss the choice of the communication protocol, which dictates the limited physical device communication radius. We chose to focus on a stable protocol that most modern devices support and the populace is most likely to use in disaster zones: Bluetooth, and more specifically Bluetooth Low Energy (BLE) [17]. We decided to focus on Bluetooth Low Energy (BLE) as it is more power efficient, over classic Basic Rate/Enhanced Data Rate (BR/EDR), as BLE conserves device battery life, it allows users to communicate over short ranges over a longer period of time than comparable communication protocols e.g. WiFi Direct [29].

Next, we discuss the bottleneck of unstructured p2p networks: the communication algorithm. Most unstructured p2p networks rely on gossip protocols to disseminate information across the network, which reduces its performance and poses a threat to resilience (cite guided research paper?). Gossip protocols usually require constant chatter between peers to reach consensus about the system state, which introduces a massive performance overhead. This performance overhead poses a risk to usablity of such systems in disaster zones, as peer devices often have limited connectivity and power supply, which together with the gossip protocol, can severely limit the peers' ability to communicate. The introduction of permissioned peers, such as high-availability bootstrapping nodes or signaling servers, improves the network's overall ability to communicate, bootstrap new communication links as well as maintain existing ones. However, those nodes pose an inherent tradeoff between performance and resilience to failure or surveillance, as relying on such nodes in disaster zones poses a risk with the infrastructure being oftentimes affected by network partitions, political censorship, or physical disruptions which halts the entire network from communicating. While such bootstrapping nodes ease the process of joining the network, discovering peers, and (potentially) disseminating and ordering state, they weaken the peers' ability to function without their presence, which is, as we show in this paper, oftentimes unnecessary. Instead of using permissioned nodes to facilitate communication by achieving total order of transactions, we suggest that using a partial ordering of transactions using a distributed Directed Acyclic Graph Conflict Free Replicated Data Types (DAG-CRDT), in which each node contains blocks of data, as well as edges which reference previous blocks, lends itself well to storing persistent communication data needed for social networking. This knowledge base, also known as the block frontier, provides proof to the local state of the node which created it, and iteratively ordering it on each participants' local device using a total order, achieves an eventual total order without extra communication between the peers [30].

In this research paper, we deal mainly with the following three axes of distributed systems: security, decentralization, and resilience. In the first axis, security, we discuss the security of an open distributed system, its permission model, anonymity, confidentiality, and encryption. Following the security axis, we explore decentralization as a way to circumvent centralized control and avoid single points of failure. Several design decisions in our system architecture influence the system's decentralization: a pure p2p system completely resides on the peer devices; while this approach highlights communication which is independent from any existing networking architecture, it introduces limitations on the range of communication, as it only allows for communication based on proximity due to the physical device constraints. Finally, we discuss resilience, which in our context refers to network survivability under duress. These communication issues include Byzantine behavior, which is independent of the communication. These issues include reachability, bandwidth limitations, and coverage loss.

This paper proposes a system architecture for resilient unstructured p2p networks, based on a wireless dual communication stack that leverages both a short-range city-scale relation as well as as Internet-scale relation, when available, to facilitate effective communication. For this, in section 2, we present the p2p communication protocols we used: BLE and UDX 2.1, as well as related works which lay at the foundation of this research paper: Peer-to-Peer Opportunistic Networking 2.2, Routing in Opportunistic Networking 2.3, Decentralized Social Networks (DSNs) 2.4, and finally, Grassroots Social Networks, and what sets them apart from DSNs 2.5. Finally, the State-of-the-Art chapter positions this research work contrastingly.

To formalize the setting, goals, and environment for this research paper, section 3 will introduce our basic environment assumptions, an analysis of the concrete problem at hand, and the design goals of the system. Elaborating on the approach introduced in chapter 3, section 4 introduces the system architecture designed to support a permissionless, unstructured p2p social network, emphasizing the base algorithms, system components, and different communication layers. The 5 section introduces the testbed we used to evaluate the performance of the system architecture introduced in 4. Next, chapter 6 evaluates the results of the experiment conducted on the testbed, which proves the possibility of our design. Chapter 7 discusses the implications of the results introduced in the previous chapter, and finally, chapter 8 concludes the paper.

2. State-of-the-Art

The State-of-the-Art section presents the papers that lay at the foundation at this research work, and positions it adverseraly. The sections in this chapter range will provide an overview over the communication mediums that we considered for this work: Bluetooth Classic, Bluetooth Low Energy, Bluetooth Mesh, Wi-Fi ad-hoc and Direct, and others. Next, this section continues with elaborating the networking assumptions that make this work possible, primarily discussing the difference between traditional packet-switching and the Store-Carry-Forward paradigm, that enables Delay-Tolerant Networking, while also presenting similar systems that have been developed using this paradigm. Followingly, we present main approaches that enable packet propagation in Delay Tolerant Networks (DTNs), explain each one of them, and show which one we used. Then, we present leading Decentralized Social Networks, and discuss whether they could have been used in our scenarios, showing that they actually cannot, and finally present the pinnacle: Grassroots Social Networks. Grassroots Social Networks lay at the core of this research work, and the networking layer presented here is a core work that lays at its foundation.

2.1. Peer-to-peer Communication Technologies

To create the peer-to-peer communication platform, different technologies were evaluated based on the following criteria:

- **C1** Spread of the protocol: As a user-facing tool, the communication medium had to be present in virtually all smartphones used today; thus, technologies that are supported by a handful of flagship smartphones, while being an interesting point for future work, are out of the question for this work.
- **C2** Low power: the communication protocol should not use a high amount of power to propagate packets
- **C3** Target: the communication protocol should be able to carry text messages efficiently, so a text message should be received within a short time in close range (e.g. 200KB in 1 second).
- **C4** Mobility: protocols neede to support a high node mobility stemming from the mobility patterns of the users
- **C5** Infrastructure: there can be no external infrastructure necessary for the normal operation of the technology

2.1.1. Wi-Fi

To relay information directly between peer smartphones, the omnipresent Wi-Fi protocol was first investigated. Propelled by the Wi-Fi Alliance, WLAN is the wireless 802.11 protocol which enables most of the world's computers to communicate. Together with cellular-based mobile networks like 5G, home and commercial routers provide wireless radio access networks for mobile devices to connect to the Internet backbone through a multitude of devices: antennas (Radio Units), computers for signal processing (Distributed Units), base stations (Centralized Units), and more [31]. For the standard Internet-backbone to function, all the aforementioned infrastructure units need to have a constant stream of power, an unblocked radio spectrum, and operational staff. In emergency scenarios such as natural disasters, wars, or politically-motivated violence against the freedom of speech through censorship, media control, or monitoring, the aforementioned infrastructure might not be available for usage. As alternatives to the classical Wi-Fi WLAN technology to facilitate DTNs, we evaluated the following p2p communication media based on the aforementioned common criteria: Wifi Adhoc, Wifi Direct, Wifi Aware, Bluetooth BDR, Bluetooth Low Energy. The Wi-Fi Ad-Hoc mode was the first design to adapt the WLAN standard to direct peer-to-peer wireless communication. However, it suffered from several notable architectural shortcomings, including inefficient power management, a lack of robust security mechanisms, and the requirement for devices to manually decide whether to act as the access point or a client [32]. In the Wi-Fi Ad-Hoc mode, idle smartphones suffer from battery drain due to a lack of distributed power saving mode [29]. However, actively communicating smartphones had the same power cost as using a standard access point, and comparing to Bluetooth, Wi-Fi Ad-Hoc still has a vastly superior throughput rate, which is optimal for larger file transfers. In our context it is still not as useful due to **C3**: the transfer size is minimal, and we optimize for efficient battery usage through **C2**. Wi-Fi Ad-Hoc was eventually replaced by Wi-Fi Direct, which modernized peer-to-peer connectivity by adding better security and an autonomous negotiation protocol that allows connecting devices to automatically determine which one takes over the role of the access point. However, Wi-Fi Direct has two main limitations: it does not support device discovery in the background if the utilizing application is stopped or not actively running in the foreground; another significant constraint of Wi-Fi Direct on modern mobile platforms is that it is not supported by iOS devices, as Apple developed a competing standard, AirDrop [33]. The Android equivalent of AirDrop, QuickShare, is developed cooperatively by Samsung and Google, and since November 2025 supports platform interoperability between Android and iOS, and has had support transfers between Androids, Windows computers, and Chromebooks before. While similarly providing superior data rates, QuickShare, as it uses a combination of Wi-Fi Direct and Bluetooth, is still not usable for this work, as its core lies with Wi-Fi Direct, which requires more power and devices to have the using app constantly foregrounded. Wi-Fi Aware (also known as Neighbor Awareness Networking / NAN) overcomes the background limitation of Wi-Fi Direct by utilizing a highly power-efficient *discover-before-connect* paradigm. Unlike Wi-Fi Direct, Wi-Fi Aware allows devices to automatically organize into synchronized local clusters and use a publish-and-subscribe mechanism to discover nearby services and exchange lightweight messages in the background, all without establishing a formal network connection. Because it operates without maintaining an active connection state, it has a significantly lower power and memory footprint. If devices need to share large amounts of data, they can seamlessly upgrade their discovery session into a fully bi-directional, high-throughput network connection. Positioning itself as the most appropriate for our use-case, Wi-Fi Aware seems like a promising candidate; however, it was primarly intended for IoT devices [34], support for it on iOS was only recently introduced [35], and thus it was deemed not widely supported enough. Early research into ad hoc and delay-tolerant networks frequently relied on legacy 802.11b and 802.11g standards, which were highly capable for active data transfers but suffered from severe battery drain when devices were left idle [36]. In later standard, 802.11n has introduced a significantly higher throughput and a 17% improvement in energy efficiency over previous standards [29].

2.1.2. Alternatives

With Wi-Fi standards being exhaused for picking the base communication protocol, we evaluated other protocols such as Zigbee and LoRA, but very quickly forsaken due to lacking support for direct communication between mobile phones: Zigbee, LoRA, and 6LowPAN are all intended for small IoT devices to communicate among themselves or with centralized servers [34].

2.1.3. Bluetooth

A technology that is ubiquotous to all smartphones that have been released in the last 10 years is Bluetooth. Being standardized as Bluetooth 1.0 in 1999 by the Bluetooth Special Interest Group (SIG), Bluetooth was introduced to mobile phones by Ericsson in 2001 with the release of the T36, which allowed phones to replace bulky cables transfers with a wireless connection. Since then, Bluetooth has seen four major version bumps:

- Bluetooth 2.0 offered in comparison to Wi-Fi a much lower energy consumption for transferring large files, and higher resilience to signal interference from nearby transmissions.
- Bluetooth 3.0 was introduced as an upgraded specification designed to improve upon the throughput limitations of Bluetooth 2.x, doubling both send/receive throughput speeds to almost 200 KB/s, while the increasing the overall power consumption [29].
- Bluetooth 4.0 represented an elementary shift in Bluetooth technology, with the focus shifting to optimizing for ultra-low power consumption: Bluetooth Low Energy, or BLE, was introduced in this version. BLE presents until today the backbone for most modern smartphone p2p applications, due to major power-saving capabilities: because BLE supports background advertising and discovery without requiring apps to be actively foregrounded, BLE is the primary transport mechanism for decentralized networks and emergency systems like firechat, Briar, Corona-Warn-App, Twimight, BitChat, and more [26, 27, 37, 18].

We thus decided to focus on BLE. In BLE, there is a dual-role split: advertising, which is carried out by the peripheral device, and scanning, which is done by the central device. The central device controls the connection, while the peripheral provides the service. Advertising happens on a set window, in which the peripheral sends out advertisements on an interval, in which its unique identifier is published, as well as any services it supports. The BLE advertised services stem from the underlying General Attribute Protocol, or GATT. GATT services group a set of characteristics, each identified by a UUID and different read/write permissions; the data-flow between the peripheral and the central is bidirectional, with the peripheral being able to notify the central of any changes to a characteristic value as it changes, or the central can request to read the current value of a characteristic. A full guide to BLE actions can be found on PunchThrough [38]. To see the peripheral's advertisements, the central device also has to scan during the same time: the scanning is also limited to a window, during which the central listens to all incoming advertisements from surrounding devices. To discover GATT services, a device first must perform a connection, which eventually includes pairing - depending on the permissions the GATT server defined.

Table 2.1.: Evaluation of p2p communication technologies against the selection criteria.

Technology	Spread	Low power	Target throughput	Mobility	No infra-structure
Wi-Fi Ad-Hoc					
Wi-Fi Direct					
Wi-Fi Aware					
Bluetooth Classic					
Bluetooth Low Energy					

✓ fully met ∼ partially met ✗ not met

Power usage of BLE vs. Direct, NAT, Hole-punching

2.2. Opportunistic Networking Systems

(MANET BATMAN) vs. Twimight, Emergency Broadcast, COVID contact tracing, Wifi-Opp

2.3. Routing in Opportunistic Networks

Flooding, Spray-and-Pray, PRoPHET

2.4. Decentralized Social Networks

Bluesky, Mastodon, Briar, Nostr, SBB (Manyverse), Bitchat

2.5. Grassroots Social Networks

Shapiro

3. Problem Analysis

3.1. Assumptions

- Disaster struck zone: Partially down communications, BLE first, mesh
- Politically oppresive regimes: various trust levels, edge rendezvous points, p2p
- Online monitoring: encryption through Noise handshake for key exchange
- More?

3.2. Design Goals

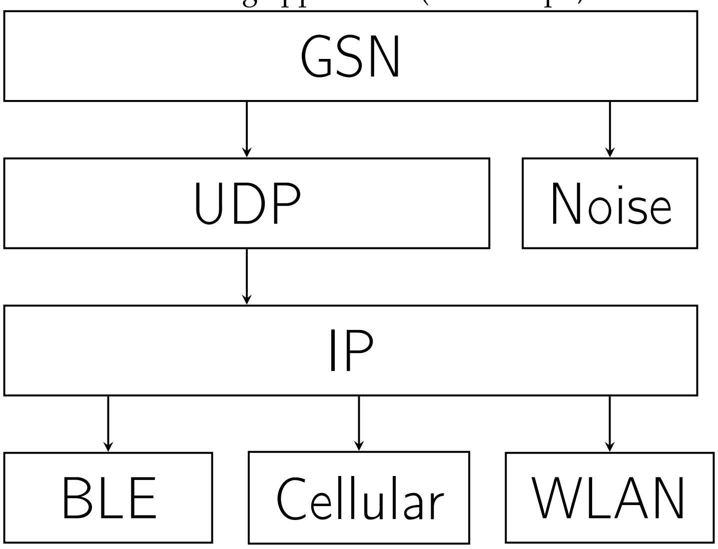
- Prefer local over remote communication
- When available, use the Internet-scale relation to relay messages over long distances
- Reduce attack interface for surveillance and monitoring as much as possible: expose the minimum amount of required information to facilitate communication
- Store-carry-forward: cache packets for unavailable recipients until a possible delivery attempt can be made
- Do not ignore existing mobile infrastructure to facilitate Internet-scale communication: edge rendezvous points and hole-punching
- Because the application may be used in various circumstances, let the user decide on various security-critical settings (i.e. use cloud RV point, friends, introduction facilitation)

4. Architecture

A demo chat app deployed on phones, which serves as the foundation to the blocklace-based future grassroots social network: argue why exactly these functions are needed.

4.1. System Architecture

The stack is organised into four layers. The lower two (transport + protocol) constitute the reusable library (‘lib/src/‘); the coordinator sits above them as the public façade; the topmost layer is the consuming application (in this repo, a demonstrator chat app under ‘lib/*.dart‘).



- Ed25519 identity
- Two transport protocols, one interface
- Message types: announce, message, friendship
- Friend vs. Stranger
- Trust Levels: cold calls
- Facilitation opt-in: edge RV points
- Always-on cloud RV points: optionally use RVs to facilitate connections
- Signaling over BT or mutual friends: hole-punching

- Simple messaging: text and media
- Blocks: ACK, NACK

5. Testbed

To evaluate the system architecture, we deployed it as a mobile application on a limited testbed of Android and iOS devices. During the deployment, usage statistics were recorded and uploaded to a secure TUM server on an opt-in basis.

5.1. CS1 — Multi-hop reach: latency & delivery vs hop count

Question. Does the mesh actually extend range beyond one radio hop, and at what latency cost?

Setup. Line topology A–B–C–D–E (allowlist or separation) $\rightarrow$ 1–4 hops.

Procedure. Fixed-size messages A$\rightarrow$B, A$\rightarrow$C, A$\rightarrow$D, A$\rightarrow$E; many repetitions; repeat with an unfragmented ($\leq$442 B) and a fragmented (multi-packet) payload.

Measures. `message dir=recv` (`relayHops`, `deliveryMethod`), `dir=delivered` (`e2eLatencyMs`), `dir=ack_timeout` (failures).

Output. Latency CDF per hop count; delivery ratio per hop count; fragmentation penalty as a second series.

5.2. CS3 — The cost of managed flooding: total mesh effort

Question. How many on-air transmissions does one delivered message cost, and how effective are the flood controls?

Setup. Same message workload run under three topologies: full mesh, line, star.

Procedure. Join `relay` records across devices on `packetId` (equals `messageId` for single-packet messages).

Measures. Total effort = origin `sent.fanout` + Σ `relay action=relayed fanout`; redundancy absorbed = `suppressed_dup` (relay side) + `message dir=dup` (recipient side); flood-control hits = `suppressed_ttl` / `suppressed_budget`.

Output. Transmissions-per-delivery vs node degree; suppression breakdown; relay load per node (the middle-node burden in the line).

5.3. CS4 — Battery: the price of relaying for strangers

Question. What does open relaying cost a node that neither sends nor receives?

Setup. Line topology; B–D relay everything, A/E are endpoints. Baselines: app idle (no traffic), app off.

Procedure. Fixed message rate for several hours per role; rotate roles across devices to cancel battery-health differences.

Measures. `device` samples (`batteryDrainPctPerHr`), `relay` records for per-node relay counts (correlate drain with effort).

Output. Drain rate per role (source / relay / sink / idle); drain vs relayed-packets-per-hour.

5.4. CS8 — Churn & real mobility: the field day

Question. Does it all hold up under real human movement?

Setup. 5 participants, one office/campus floor, ∼6 h. Zones with known radio relationships (pre-tested): Z1 office, Z2 lab (marginal reach to Z1), Z3 cafeteria (isolated).

Procedure. 15-min slot schedule per participant; workload driver runs the seeded-Poisson load throughout; traces auto-upload every 10 min.

Window	A	B	C	D	E	Produces
09–10	Z1	Z1	Z1	Z1	Z1	full-mesh baseline
10–11	Z1	Z1	Z2	Z2	Z3	stable partition; E isolated
11–12	Z1	walker	Z2	Z2	Z3	B mules Z1↔Z2 (10-min cycle)
12–13	Z3	Z3	Z3	Z1	Z1	partition flip → DTN flush
13–15		free movement				natural churn

Table 5.1.: CS8 slot schedule per participant across the field day.

Measures. Everything: `message`, `relay`, `contact`, `density`, `buffer`, `device`.

Output. Delivery-delay CDF split by direct/relayed/DTN; inter-contact CDF vs planned schedule; delivery ratio per window. **Headline criterion:** every message offered during the partition windows is delivered after the 12:00 flip.

6. Evaluation

7. Discussion

8. Conclusion

A. General Addenda

If there are several additions you want to add, but they do not fit into the thesis itself, they belong here.

A.1. Detailed Addition

Even sections are possible, but usually only used for several elements in, e.g. tables, images, etc.

B. Figures

B.1. Example 1

✓

B.2. Example 2

✗

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